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Hydrogen-rich gas as a fuel for the gas turbines: A pathway to lower CO₂ emission

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ABSTRACT

Hydrogen-rich fuel for the gas turbines can be considered as a transient way from hydrocarbon fuel to zero-carbon hydrogen fuel. In this paper, thermodynamic analysis of a combined cycle power plant (CCPP) fired with a hydrogen-rich fuel obtained via various ways is performed to understand the effect of a transient to this fuel on the CO₂ emission as well as the efficiency of a power plant. Hydrogen-rich fuel with different content of hydrogen is analyzed. Two types of hydrogen-rich fuel are considered: fuel obtained via methane dilution with hydrogen (hydrogen from internal fuel supply infrastructure) — first case; fuel obtained via steam methane reforming (on-board hydrogen production technology) — second case. The first case showed that hydrogen addition to methane non-linearly leads to a decrease in CO₂: hydrogen-rich fuel with 20% of H₂ volume fraction gives a reduction in CO₂ emission of 7.2%; 50% of H₂ gives 23.5% reduction; 75% of H₂ gives 51.1% reduction. The second case is considering hydrogen-rich fuel obtained via steam methane reforming using a renewable energy source. Hydrogen volume fraction up to 75% in hydrogen-rich fuel can be obtained after the reforming process. When 100% of methane is reformed, the reduction in CO₂ emission up to 27% can be achieved. The minimum achievable level in CO₂ emission is 75.17 g_{CO₂}/kW for an on-board hydrogen production technology via steam methane reforming which is corresponding to a hydrogen-rich fuel obtained via H₂ dilution up to 53%.

1. Introduction

The development of modern society is inextricably linked with the consumption of energy — heat, electricity, and fuel. It is recognized that the gross domestic product of most countries directly depends on the generation and consumption of electrical energy. Over the past decades, the impact of energy on the environment has become especially noticeable [1]. The consensus forecast of scientists shows that a large number of natural disasters are caused by an increase in greenhouse gas emissions, in particular, carbon dioxide because the emission of carbon dioxide is one of the main causes of global warming [2–4]. It is found that the surface temperature of the earth has increased by 0.6 °C during the twentieth century, and it is predicted to increase by 5.8 °C by 2100 [5]. An increase in the temperature of the earth's surface leads to noticeable changes in the environment, in particular in the Arctic. Melting permafrost increases natural emissions of CO₂ and other greenhouse gases, in particular methane, from peatlands, such as the peatlands of Siberia [6]. In this case, a chain reaction occurs, which is extremely difficult for people to control. Therefore, the control and reduction of CO₂ emissions is a global challenge.

In order to control and reduce CO₂ emission, most of the world (195 countries) have signed the Paris Agreement [7]. The main goal of this agreement is a decrease of the global temperature to 2 °C due to a reduction in carbon dioxide emissions up to 25% in 2030 in comparison to 2010. Moreover, according to the agreement, zero CO₂ emission must be achieved in 2070 [8]. Achieving these ambitious goals is inextricably linked to energy: electricity and heat generation, industry, transport, etc. The main reason for CO₂ emission in these sectors is hydrocarbon fuel consumption — the main source of anthropogenic CO₂ emission [9,10].

One of the obvious ways to reduce carbon dioxide emissions is to use renewable energy sources: wind, solar, hydropower, etc. [11,12]. Most developed and developing countries have adopted programs for the transition to renewable and carbon-free energy sources [13,14]. By 2020, the generation of electricity from renewable energy sources (including hydropower, solar, wind, geothermal, and biomass) is more than 50% in countries such as Norway (99%), Brazil (79%), Sweden (67%) [15]. However, several large countries still have the main share

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of electricity generation from fossil fuels: India (78%), China (69%), Japan (68%), USA (60%) [15]. Globally, the share of electricity generation from renewable sources has increased from 19% in 2000 to 28% in 2020 [16]. Obviously, despite the success in the development of renewable energy source utilization, in the coming years, the generation of electricity from fossil fuels will be the main one.

Steam and gas turbines are the traditional direction of using fossil fuel for electricity generation [17–19]. Gas turbines generated about 22% of the world's electricity in 2020 [20]. The world's total gas turbine capacity is about 1650 GW. Gas turbines mostly use natural gas, consisting mainly of methane, as a fuel to generate electricity. In addition, there is a wide experience of using other fossil fuels in gas turbines: diesel, light oil, gasoline [21–23]. In addition, the exploitability of gas turbines can be more than 20 years. Therefore, it is obvious that the generation of electricity using gas turbines in the coming decades will have a significant share in the global energy balance. At the same time, the issues of reducing CO₂ emissions will continue to be important for this direction of electricity generation.

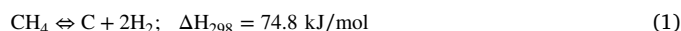
What are the ways to reduce CO₂ emissions in gas turbines? There are two directions for CO₂ emission reduction in gas turbines: post-combustion [24,25] and pre-combustion [26,27]. The post-combustion way is based on the capture of CO₂ from flue gases and utilization of captured CO₂. Post-combustion CO₂ capture is one of the promising ways because this method can be applied in the existing gas turbines [28,29]. The main methods of post-combustion CO₂ capture can be highlighted in the present time: chemical absorption [30], adsorption [31], membranes [32], photochemical capture [33], etc. The first three methods are based on CO₂ utilization and storage. Captured carbon dioxide is compressed and transported to the consumers in the industry or for underground storage in geological strata [34,35]. The photochemical capture is the most environmentally friendly because CO₂ is used for biomass production, for example, algae [36]. The main idea of this method is based on CO₂ solving in water and used by algae for photosynthesis. The total CO₂ emission from electricity generation is about 1500 million metric tons of CO₂ equivalent (as of 2020) [37]. Therefore, global post-combustion CO₂ capture cannot be applied due to huge industrial CO₂ emissions. However, pointwise this method can be used successfully in some cases. Moreover, when discussing CO₂ capture issues, it is important to highlight one of the promising ways of achieving CO₂-neutrality: production and use of electro-fuels (e-fuels) [38].

Pre-combustion decarbonization of fuel for gas turbines is based on the concept of low- or zero-carbon fuel used for electricity generation. Two main ways of pre-combustion decarbonization of a fuel for gas turbines can be distinguished: carbon extraction and capture from a hydrocarbon fuel [39]; utilization of low- or zero-carbon fuel (non-carbon fuel) [40–42].

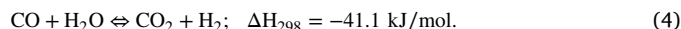
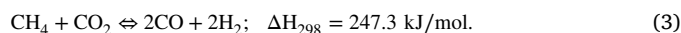
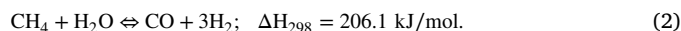
The first way is applied for the use of hydrocarbon fuel. Natural gas, in particular, methane is the main fuel. Thus, carbon extraction and capture from hydrocarbon fuel before combustion can be considered for CH₄ as an example. Methane is consisting of one atom of carbon and 4 atoms of hydrogen. The different mechanisms of carbon extraction and capture can be used. However, these mechanisms can be divided into two groups. The first one is methane cracking to obtain solid carbon and it captures in a solid form [43–45]. The second one is methane reforming to obtain either carbon dioxide or carbon monoxide and it captures in a gaseous form [46–48].

The concept of solid carbon capture is based on methane cracking reaction (1). As can be seen below, the methane cracking reaction is endothermic and external heat must be supplied to occur this reaction. Various heat sources can be utilized for this reaction: renewable heat sources (e.g. solar energy), waste heat after a gas turbine, etc. [49,50]. This way is one of the most industrially applicable due to the simplicity of carbon capture in a solid form. After capture, solid carbon can be successfully disposed of in mines or geological formation. In addition,

solid carbon is a useful material that can be widely re-used in various industrial applications [45,51,52].



The second mechanism of carbon capture from methane is methane reforming with either steam (2) or carbon dioxide (3). In this case, methane is reacted with steam or carbon dioxide to produce the mixture of hydrogen and carbon monoxide [53,54]. In some cases carbon monoxide can be converted into the mixture of hydrogen and carbon dioxide (water-gas shift reaction (4)) [55,56].



The products of reforming can be separated via various techniques: membrane [57–59], sorption enhanced [60,61], etc. The membrane technology is based on penetration of H₂ through a membrane. The membrane reactors are a variation of methane reformers. In this installation, hydrogen is seeping through a membrane and two streams are taking place at the outlet — pure hydrogen and CO₂-rich gas mixture. The different types of membranes can be applied for this goal: Pt-, Ag-based, etc. [62,63].

Sorption-enhanced steam methane reforming is based on CO₂ capture and production of pure hydrogen. This method is almost similar to CO₂ capture from the flue gases — post-combustion CO₂ capture. In-situ carbon dioxide capture is taking place in a high-temperature solid sorbent, for example, CaO-based sorbent [64,65]. The chemical reaction of carbon dioxide and CaO-based sorbent gives carbonate calcium in a solid form that can be efficiently stored.

The second way of pre-combustion decarbonization is the use of low- or zero-carbon fuels: ammonia [66], ammonia–methane mixture [67], hydrogen-rich gas [68], hydrogen [69], etc. If such fuels contain no carbon, no carbon dioxide emission will be observed. Last years, ammonia as a fuel finds the wide interest of engineers and scientists. Mitsubishi Power reported about the plans to build a 40-MW gas turbine fueled with 100% ammonia [70]. In this case, ammonia is considered an energy carrier for a gas turbine. The ammonia-fired gas turbines are an example of carbon-free power generation systems [71]. The use of ammonia as a fuel meets different challenges, the main of which are the following: NO_x emission and different properties of the combustion process. The addition of methane to ammonia makes the methane–ammonia mixture close to the combustion performances of pure methane [67].

Hydrogen as a fuel is one of the perspective ways to the zero-carbon emission gas turbines. The products of hydrogen combustion are mainly water and a small volume fraction of NO_x when air is used for combustion. The perspectives of hydrogen as a fuel is widely discussed [72–74]. However, most investigators highlighted the following challenges in the use of hydrogen as a fuel: storage and transportation, combustion, and effect on mechanical properties [75–77]. Therefore, while ammonia can fully replace methane as a fuel, hydrogen utilization as a methane change is meeting with challenges.

One of the ways to a smooth transition from the use of methane to hydrogen utilization as fuel is the addition of hydrogen to methane to obtain the hydrogen–methane blend [78,79]. In this case, hydrogen-rich fuel is used as fuel. There are several examples of hydrogen-rich natural gas utilization on existing installed gas turbines. General Electric reported that they have about thirty gas turbines that use fuels with at least 50% hydrogen (by volume) [80]. Existing fuel supply infrastructure with slight modernization is used for such turbines. The main goal of hydrogen addition to methane is getting the same output power with lower CO₂ emission. Another way to obtain hydrogen-rich gas is the utilization of products of hydrocarbon fuel reforming [81].

For example, methane reforming reaction (2) produces the hydrogen-rich gas with 75% volume fraction of H_2 . The composition of the reforming products is strongly depending on operational and constructive parameters of the reforming process: temperature, catalyst, pressure, residence time, etc. The reforming reactions are endothermic and external heat must be supplied to the reaction space. Various heat sources can be used for this goal, for example, renewable solar energy [82,83], exhaust waste heat [84,85], etc. Hydrogen-rich gas after the reforming process has some advantages in comparison to hydrogen-rich gas after the addition of hydrogen to methane. One of them is a possibility to produce hydrogen-rich gas before using it as a fuel without a need of hydrogen storage. In this case, the concept of on-board hydrogen production is taking place.

The use of hydrogen-rich gas as fuel gives a smooth transition from hydrocarbon fuel to low- or zero-carbon fuels. A decarbonization concept based on hydrogen-rich fuel can be employed on existing gas turbines without global modernization of the schematic diagrams [86]. The scheme of a gas turbine fueled with methane (natural gas) would remain similar to the combustion of hydrogen-rich fuel. In this paper, the use of hydrogen-rich gas as a fuel for gas turbines is analyzed. Two types of hydrogen-rich fuels are considered: methane/hydrogen gas mixture and the products of hydrocarbon fuel reforming. The thermodynamic analysis of a gas turbine fueled with hydrogen-rich gas of various compositions is performed for the different schematic diagrams: methane dilution with hydrogen and on-board hydrogen production via fuel reforming. The effect of hydrogen addition and fuel reforming on CO_2 emission is analyzed in the combined cycle power plant. Moreover, the effect of the transition to hydrogen-rich fuel from methane is determined for lower and higher heating value efficiency. The mass flow rates through gas turbines with various compositions of hydrogen-rich fuels are analyzed.

2. Methane dilution with hydrogen

One of obvious ways to transit from hydrocarbon fuels (e.g. natural gas) to low carbon fuels (e.g. hydrogen-rich gas) is the addition of hydrogen to hydrocarbon fuel. In this case, the share of hydrogen in fuel is increasing, and the share of hydrocarbon fuel is decreasing. Natural gas dilution with pure hydrogen is one of the simplest ways to use the hydrogen-rich fuel in the existing gas turbines [78,87,88]. In April 2022, Long Ridge Energy Terminal and General Electric reported the results of successful utilization of methane–hydrogen mixture in the combined cycle power plant [89]. The tests were performed to show the capability of the use of natural gas diluted with hydrogen. During the tests on GE7HA.02 gas turbine, the engineers used a mixture of hydrogen and methane in different proportions. Hydrogen-rich gas with 5%–25% of pure hydrogen was used as a fuel. Other sets of tests were performed to provide detailed information about the possibility of the use of the hydrogen-rich fuel with 25%–50% of hydrogen.

The tests on combined cycle power plants in Long Ridge Energy Terminal showed that hydrogen-rich gas can be considered a more climate-friendly fuel for power generation in comparison with traditional hydrocarbon fuel, e.g. natural gas, because up to 20% of reduction in CO_2 emission was observed in the existing power plant. These tests were one of the first steps to getting to low-carbon energy. According to General Electric's report, hydrogen for dilution was produced on-site before utilization, and the volume fraction of hydrogen plans to be increased up to 100%. Hydrogen-rich fuel obtained via natural gas dilution with hydrogen can be used in the existing power plants without sufficient modernization. The schematic diagram of the classical combined cycle will only need the addition of hydrogen supply infrastructure.

It should be noted that the addition of hydrogen to methane does not reduce CO_2 emissions if the hydrogen is produced from hydrocarbon fuels without CO_2 capture. The ways of hydrogen production are

classified by color differentiation. Table 1 presents the classification of the different ways of hydrogen production.

Currently, there is no reliable information on the use of 100% hydrogen as a fuel in the existing high-power gas turbines in the industry. Most researchers believe that the use of hydrogen-enriched gas will be the main direction in the transition to low-carbon energy. Chiesa et al. considered the possibility of the use of hydrogen as a fuel in gas turbines [99]. The authors paid attention to the aspects of switching from natural gas to hydrogen. They considered various ways to transit from natural gas to hydrogen in the gas turbines. The results showed that the use of pure hydrogen as fuel leads to an increase in NO_x emissions. The main reason for higher NO_x emission is higher combustion temperature of hydrogen comparing with the combustion temperature if methane. Therefore, the authors considered a gas fuel where hydrogen is diluted with steam and nitrogen. Switching from natural gas to hydrogen leads to a lower mass flow rate of the working fluid. Moreover, the composition of the working fluid is different with a higher steam fraction. In the conclusion, the authors reported that 100% hydrogen as a fuel will meet engineering problems in existing gas turbines. They focused on the behavior of the gas turbine, by considering the effects of the variation of volume flow rates and of thermophysical properties, related to switching from natural gas to hydrogen. Therefore, the authors concluded that hydrogen diluted with inert gases such as nitrogen and steam can be used as a fuel in gas turbines.

In 2007, the U.S. Department of Energy (DOE) developed Advanced Hydrogen Turbine Development Program [100]. The program was focused to develop a combined power plant based on SGT6-6000G gas turbine fueled with hydrogen. The engineers from Siemens showed that to reduce NO_x emission, pure hydrogen should be diluted with other gases. They considered synthesis gas as a potential hydrogen-rich fuel for gas turbines to get an appropriate level of NO_x emission.

Oberg et al. recently performed a techno-economic study of gas turbines fueled with hydrogen for the low-carbon dioxide emissions systems for electricity generation [101]. The authors analyzed the blends of hydrogen and biomethane as fuel for gas turbines. They showed that utilization of hydrogen produced in the solar or wind energy systems can be efficient with a combination of renewable methane (biomethane) even if a hydrogen cost is up to 5 €/kg H_2 .

Sun et al. investigated the combustion and emission characteristics of methane-hydrogen mixture as a fuel of the gas turbines [102]. The authors paid attention to the study of NO_x emission. They showed that the addition of H_2 leads to an increase in NO_x emission due to an increase in the combustion temperature. However, the addition of hydrogen leads to decreasing in CO_2 emission.

Most of the above-mentioned authors highlighted the different properties of hydrogen and methane: the heat of combustion, flame speed, etc. To understand the calorimetric properties of the hydrogen–methane mixture, various compositions were analyzed. Fig. 1 shows the calorimetric characteristics of hydrogen-rich gas and CO_2 emission: (a) H_2 mass fraction and mass of CO_2 per 1 kg of hydrogen-rich fuel as a function of H_2 volume fraction in fuel; (b) Higher (HHV) and lower (LHV) heating value of 1 kg of hydrogen-rich fuel and mass of hydrogen-rich fuel to obtain HHV (55.5 MJ/kg) and LHV (50.0 MJ/kg) of 1 kg of methane.

The density of hydrogen is about 8 times less than the density of methane. As it can be seen from Fig. 1a, the mass fraction of hydrogen in H_2 - CH_4 -mixture is increasing non-linearly. 50% of H_2 volume fraction gives only 11% of H_2 mass fraction. As a consequence, carbon dioxide emission per 1 kg of hydrogen-rich fuel is also non-linear depending on H_2 volume fraction. The mass of CO_2 per 1 kg of hydrogen-rich fuel with 50%(vol.) of H_2 is 11% less than the mass of CO_2 for pure methane combustion. However, the heat of combustion of hydrogen per 1 kg is more than 2 times higher than the heat of combustion of methane. As a result of which the mass of CO_2 per 1 MJ of the combustion heat is 42 g CO_2 for hydrogen-rich fuel with 50%

Table 1

A color-based classification of the hydrogen production ways [90].

Color of H ₂	Process	Chemical reaction	Remark
Black (Gray)	Natural gas reforming	$\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$	No CO ₂ capture
Blue	Natural gas reforming	$\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$	With CO ₂ capture and storage [47,91]
Green	Water electrolysis	$2\text{H}_2\text{O} \rightleftharpoons 2\text{H}_2 + \text{O}_2$	Renewable power [92,93]
Red (pink)	Water electrolysis	$2\text{H}_2\text{O} \rightleftharpoons 2\text{H}_2 + \text{O}_2$	Nuclear power [94,95]
Turquoise	Methane pyrolysis	$\text{CH}_4 \rightleftharpoons \text{C} + 2\text{H}_2$	Use of carbon as a by-product [51,96]
White	Biomass gasification	$\text{C}_x\text{H}_y\text{O}_z + \text{O}_2 \rightleftharpoons 0.5y\text{H}_2 + x\text{CO}_2$	100% of biomass as feed stock [97,98]

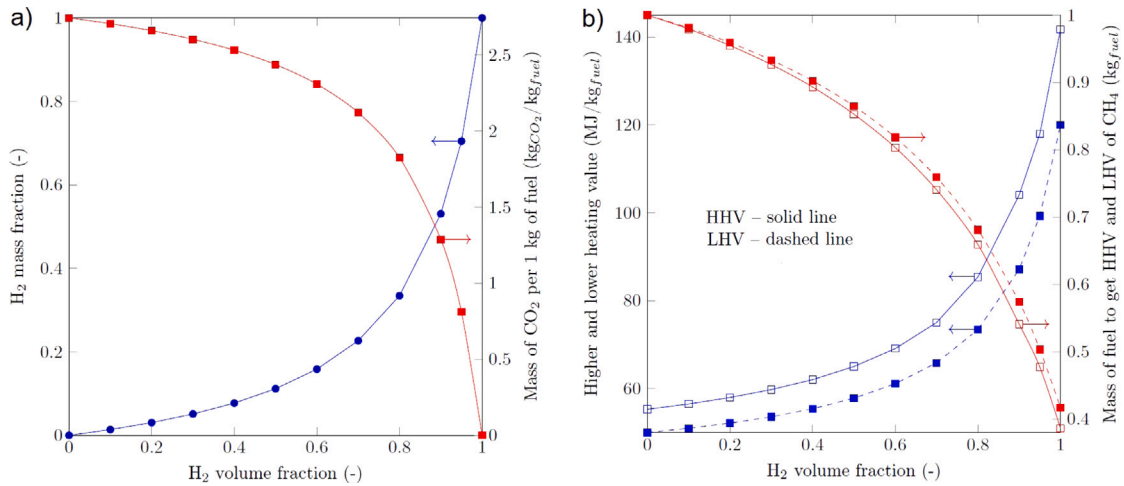


Fig. 1. The calorimetric characteristics of hydrogen-rich gas and CO₂ emission: (a) H₂ mass fraction and mass of CO₂ per 1 kg of hydrogen-rich fuel as a function of H₂ volume fraction in fuel; (b) Higher (HHV) and lower (LHV) heating value of 1 kg of hydrogen-rich fuel and mass of hydrogen-rich fuel to obtain HHV (55.5 MJ/kg) and LHV (50.0 MJ/kg) of 1 kg of methane.

(vol.) of H₂, while the mass of CO₂ per 1 MJ of the combustion heat is 54 gCO₂ for pure methane. In this case, a decrease in CO₂ emission is about 22%. The results from Fig. 1a illustrates that an increase in H₂ volume fraction above 50% will lead to a significant decrease in CO₂ emission. At the same time, an increase in H₂ volume fraction up to 50% does not play an important role in decreasing CO₂ emission.

Methane dilution with hydrogen leads to changes in the calorimetric performances of hydrogen-rich fuel. H₂ addition leads to an increase in volume and mass fraction of H₂O in the combustion products. Therefore, the ratio of the higher heating value (HHV) and lower heating value (LHV) is increasing with an increase in H₂ volume fraction. This ratio for pure methane is about 1.12, while for pure hydrogen is about 1.18. This fact is illustrated in Fig. 1b. Fig. 1b also shows that the mass of hydrogen-rich to obtain HHV or LHV of methane is decreasing with increasing H₂ volume fraction. To generate 50 MJ of heat, 0.85 kg of hydrogen-rich fuel with 50% (vol.) of H₂ should be burned in comparison to 1 kg of pure methane.

2.1. Methodology

H₂ addition to methane leads not only to changes in combustion characteristics and CO₂ emission but also to changes in mass and heat flows in the gas turbine cycle. To understand the effect of H₂ addition to methane, the thermodynamic analysis of a combined cycle power plant was performed via Aspen HYSYS. The combined cycle power plant is based on the gas turbine 7HA.02 produced by General Electric with a power of 330.000 kW. The gas turbines of 7HA-series are installed at Long Ridge Energy Terminal where the combustion of hydrogen-rich fuel was investigated. Exhaust heat after a gas turbine is used to produce steam for a steam turbine cycle. The parameters of the combined cycle power plant are presented in Table 2.

The computational scheme of the combined cycle power plant fired with hydrogen-rich fuel is shown in Fig. 2. In this scheme, the turbine inlet temperature (1400 °C) was set as a constant for all investigated

Table 2

The parameters of combines cycle power plant with the gas turbine 7HA.02.

Parameter	Value
Gas turbine power	330.000 kW
Gas turbine inlet temperature (TIT)	1400 °C
Compression/expansion ratio	21.5
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Steam temperature at steam turbine inlet	400 °C
Steam pressure	40 bar
Condenser pressure	0.06 bar
Exhaust gas temperature after steam generator	100 °C

cases (from 0 to 100% of hydrogen volume fraction). The power of the gas turbine was also set as a constant (330 MW) for all cases. The turbine inlet temperature was regulated by an air mass flow rate with the help of *adjust* procedure (ADJ-1). The temperature of exhaust gas after a steam generator was set to 100 °C for all cases.

2.2. Results and discussion

To understand the effect of hydrogen addition to methane, the mass of carbon dioxide per 1 kW of power output has been determined. Fig. 3 shows the mass of CO₂ per 1 kW of a power output of the analyzed combined cycle power plant and reduction in CO₂ emission as a function of hydrogen volume fraction in a hydrogen-rich fuel.

Fig. 3 depicts that the addition of hydrogen to methane is decreasing in the mass of CO₂ per 1 kW. However, this decrease is not linear. Utilization of 100% methane as fuel gives about 96 grams of CO₂ per 1 kW of the power output. Hydrogen-rich fuel with a hydrogen-to-methane ratio of 1:1 (CH₄ — 50%; H₂ — 50%) produces about 73 gCO₂/kW. Therefore, the first 50% of hydrogen volume fraction gives a decrease of 23 gCO₂/kW. At the same time, an increase in hydrogen volume fraction from 50% to 75% and from 90% to 100% gives also

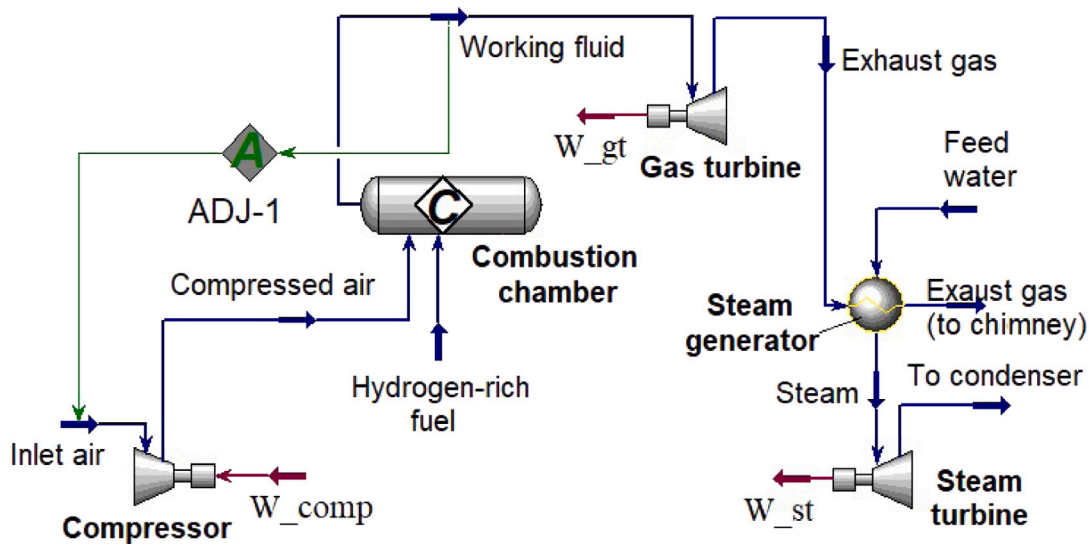


Fig. 2. A computational scheme of a combined cycle power plant fired with hydrogen-rich fuel.

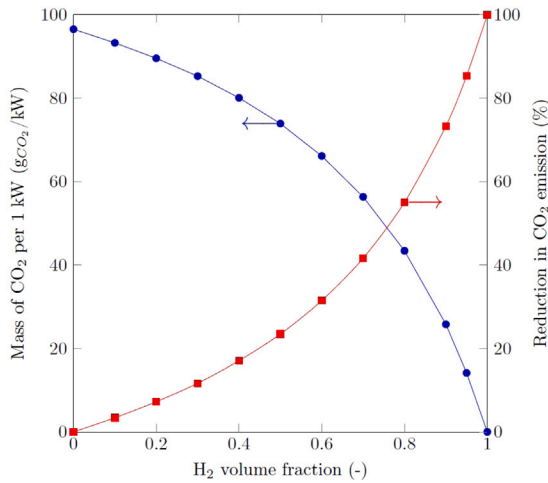


Fig. 3. Mass of CO₂ per 1 kW of power output and reduction in CO₂ emission as a function of hydrogen volume fraction in a hydrogen-rich fuel.

a decrease of 23 gCO₂/kW. Thus, it can be concluded that an increase in H₂ volume fraction above 50% will lead to a significant decrease in CO₂ emission, while the effect of methane dilution with hydrogen for the low H₂ fraction does not have a notable effect on CO₂ emission.

Fig. 3 (right side) illustrates that the addition of hydrogen to methane to obtain a hydrogen-rich fuel for gas turbines has no one-to-one ratio between H₂ volume fraction and the reduction in CO₂ emission. For example, an increase in a hydrogen volume fraction from 0 to 50% of leads to 23% of the CO₂ emission reduction. In addition, to obtain 50% reduction in carbon dioxide emission, a hydrogen-rich fuel with more than 75% of hydrogen should be used.

Full or partial replacement of methane by hydrogen leads to a change in a mass flow rate of a hydrogen-rich fuel as well as in a mass flow rate of a working fluid in gas turbines. Fig. 4 shows the effect of H₂ volume fraction on a mass flow rate of a hydrogen-rich fuel and a working fluid (a) and the performance of a combined cycle power plant (b).

Fig. 4a shows that an increase in H₂ volume fraction in a hydrogen-rich fuel leads to a decrease in a mass flow rate of fuel: 56400 kg/h for H₂=0%; 48400 kg/h for H₂=50% (16.8% less than for H₂=0%); 22920 kg/h for H₂=100% (60.4% less than for H₂=0%). Fig. 4a illustrates that a mass flow rate of a hydrogen-rich fuel is significantly

decreasing. However, the volume flow rate is increasing because the combustion heat per 1 m³ of H₂ is approximately three times less than the combustion heat of m³ of CH₄. Fig. 4a depicts that the existing fuel supply infrastructure of the gas turbines can be used at a low H₂ volume fraction. However, the fuel supply infrastructure should be adopted for hydrogen-rich fuel at a high H₂ volume fraction hydrogen has different properties than methane. Operating a gas turbine on a hydrogen-rich fuel may require changes to combustion, fuel, and plant safety systems of the power plants.

Despite a significant change in hydrogen-rich fuel consumption with an increase in H₂ volume fraction, the mass flow rate of working fluid is slightly changing. An increase in H₂ volume fraction from 0 to 50% leads to a decrease in a mass flow rate of working fluid of 1.7%; from 0 to 100% of 6.4%. Therefore, the mass flow rate of working fluid through a gas turbine is insignificantly changing and the impact on the flow rate through the flow region of the gas turbine will be minimal. The main reason is the mass flow rate of air in a compressor. The average mass flow rate of air is about 2100000 kg/h and a mass fraction of fuel is minimal in comparison with a mass fraction of air.

Methane dilution with hydrogen leads to a change in the performance of a combined cycle power plant. The effect of H₂ volume fraction in a hydrogen-rich fuel on the efficiency of a combined cycle power plant is shown in Fig. 4b. The lower and higher heating value efficiency is determined from the following expressions [103]:

$$\eta_{LHV} = \frac{N_{CCPP}}{(\dot{m}_{H_2} \cdot LHV_{H_2} + \dot{m}_{CH_4} \cdot LHV_{CH_4})}; \quad (5)$$

$$\eta_{HHV} = \frac{N_{CCPP}}{(\dot{m}_{H_2} \cdot HHV_{H_2} + \dot{m}_{CH_4} \cdot HHV_{CH_4})};$$

where N_{CCPP} is power output of combined cycle power plant; $\dot{m}_{CH_4}$, $\dot{m}_{H_2}$ is a mass of methane and hydrogen in a hydrogen-rich fuel; LHV_i , HHV_i is lower and higher heating value of i th gas component.

Fig. 4b shows that an increase in H₂ volume fraction leads to a slight increase in LHV efficiency. The difference between LHV efficiency at 0 and 100% of H₂ is approximately 0.7%. However, an increase in H₂ volume fraction leads to a notable decrease in HHV efficiency (the difference is more than 2.5%). The main reason for this finding is a change in the composition of a working fluid. H₂ combustion produces only H₂O while CH₄ combustion produces a mixture of H₂O and CO₂. An increase in H₂ volume fraction leads to an increase in H₂O molar fraction in a working fluid. Therefore, losses with the latent heat of condensation are increasing. The change in a working fluid composition

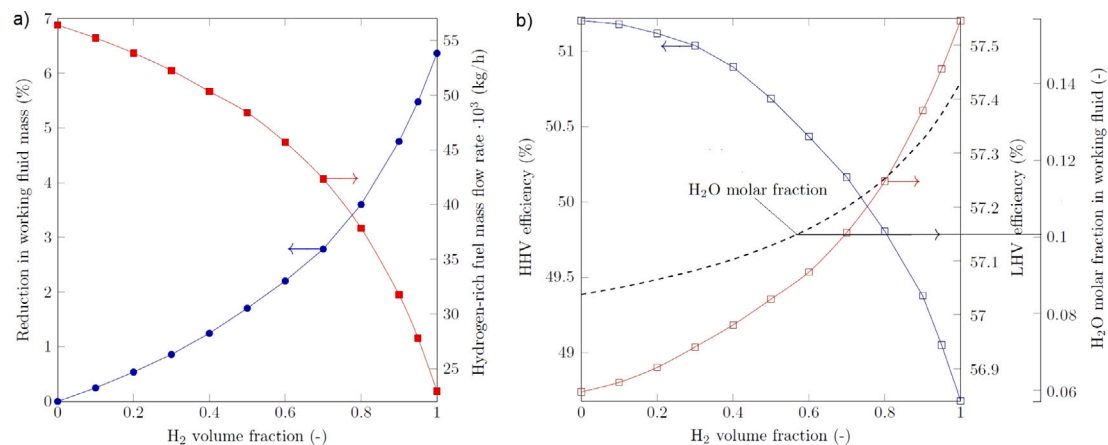


Fig. 4. Hydrogen-rich fuel mass flow rate and reduction in a working fluid mass flow rate as a function of H₂ volume fraction (a) and the effect of H₂ volume fraction on HHV and LHV efficiency of a combined cycle power plant as well as on H₂O molar fraction in a working fluid (b).

may be the reason for the installation of the condenser heat exchangers. In this case, the utilization of hydrogen-rich fuel will have an additional positive effect on the overall efficiency of the combined cycle power plants.

Another important issue that is necessary to briefly discuss but which is outside of this study is NO_x emission. Widely known that the combustion temperature of H₂ is higher than the combustion temperature of CH₄. This fact leads to increasing in NO_x formation in a combustion chamber. Some papers showed that the combustion of 1:1 of hydrogen-to-methane mixture leads to an increase in NO_x emission up to 35% in comparison with the combustion of fuel with CH₄ of 100%. An investigation of NO_x formation for various H₂ volume fractions will be presented in the future author's papers.

3. On-board hydrogen-rich fuel production

The schematic diagram of a gas turbine for the combustion of methane will be almost unchanged for the combustion of a hydrogen-rich fuel obtained via methane dilution with hydrogen. However, fuel supply infrastructure should be significantly changed because a hydrogen supply system is different from a methane supply system. The main reason for this issue is the high pressure of hydrogen (up to 350 bar) in a fuel supply system [104,105]. Moreover, the pressure of hydrogen for storage is more than 350 bar [106]. Therefore, the hydrogen storage and supply systems will require high-cost equipment and notable energy losses for compression.

One of the ways to avoid a significant change in a fuel supply system is the production of hydrogen-rich fuel before utilization in a gas turbine. This way of hydrogen-rich fuel production is named on-board hydrogen-rich fuel production or on-board hydrogen production (OBHP) technology [107]. As mentioned above, about 90%–95% of hydrogen is produced from hydrocarbon fuel [90,108,109]. The main concept of OBHP technology is based on hydrogen-rich fuel production from hydrocarbon fuel. Usually, the process of hydrogen production from hydrocarbon fuel is endothermic and for this purpose, a renewable energy source can be used, for example, solar energy.

The idea of fuel reforming before combustion was widely discussed for application in internal combustion engines (ICE) [110,111]. The product of reforming is a synthesis gas with a high volume fraction of hydrogen. And then, this hydrogen-rich fuel is used as a fuel for ICE. The most of authors proposed to use exhaust heat to compensate for the endothermic effect of the reforming process [112,113].

Hydrocarbon fuel reforming before combustion is also widely discussed for the application in gas turbines [114–116]. In these gas turbines, exhaust heat after a gas turbine is considered as the main heat source to cover the endothermic effect of the reforming process. The

gas turbines where exhaust heat is used for the endothermic transformation of initial hydrogen fuel are called “chemically recuperated gas turbines” (CRGT) [117].

In the last years, natural gas reforming before combustion is a widely discussed issue. Thermochemical exhaust heat recuperation is an example of such a way of fuel utilization [81,86]. Exhaust heat after a gas turbine is used for methane reforming with steam, and then the products of reforming with a high hydrogen mole fraction are used as a fuel in the gas turbine unit [114]. Sadeghi et al. [118] performed a comparison of the organic Rankine cycle system with CRGT based on steam methane reforming. Su et al. [119] proposed a biogas-fired cogeneration system based on CRGT cycle. Moreover, the authors reported that in CRGT a hydrogen-rich gas is used as a fuel in a combustion chamber. Lui et al. showed the results of the study of the combustion characteristics in CRGT [120] and reported that CRGT can be considered as ultralow NO_x emission technology.

There is a set of paper where methanol reforming is considered as a way for hydrogen-rich production with heat source from exhaust gases. Pan et al. reported the results of an investigation of thermochemical recuperation in a methanol-fueled gas turbine [121]. Pettersson and Sjöström provided the review on the use of the products of methanol decomposition in the gas turbine units through exhaust heat recuperation [122]. Moreover, Tola and Lonis presented a paper on the use of bio-methanol in a gas turbine with thermochemical recuperation [123]. The main attention was paid to the fact that renewable methanol can be efficiently used for reforming in gas turbines and then the reforming products are used as a fuel. In addition, methanol is the excellent electro-fuel that may be produced through CO₂ capture [124,125]. Therefore, achieving CO₂-neutrality via the use of methanol reforming is a promising case.

One of the main benefit of fuel reforming before combustion is a possibility to use the renewable energy sources for covering of the endothermic effect. Bianchini et al. [126] proposed to use solar energy for methane reforming before a combustion chamber of a gas turbine unit. The authors reported that an integrated steam methane reformer heated by solar energy, and a gas turbine make it possible to achieve up to 20% of initial methane saving. The results were obvious because solar energy replaced a part of heat supplied after methane combustion.

Moreover, Li et al. proposed a solar combined-cycle power system with methane membrane reforming as a way to get low carbon dioxide emissions [127]. In this system, the solar fields were integrated with a combined cycle power plant. Solar energy is utilized for methane reforming to produce hydrogen-rich fuel in a methane membrane reformer. The authors showed that in the proposed system is achieving of 31.2% in a fossil fuel saving in comparison to conventional combined cycle power plants.

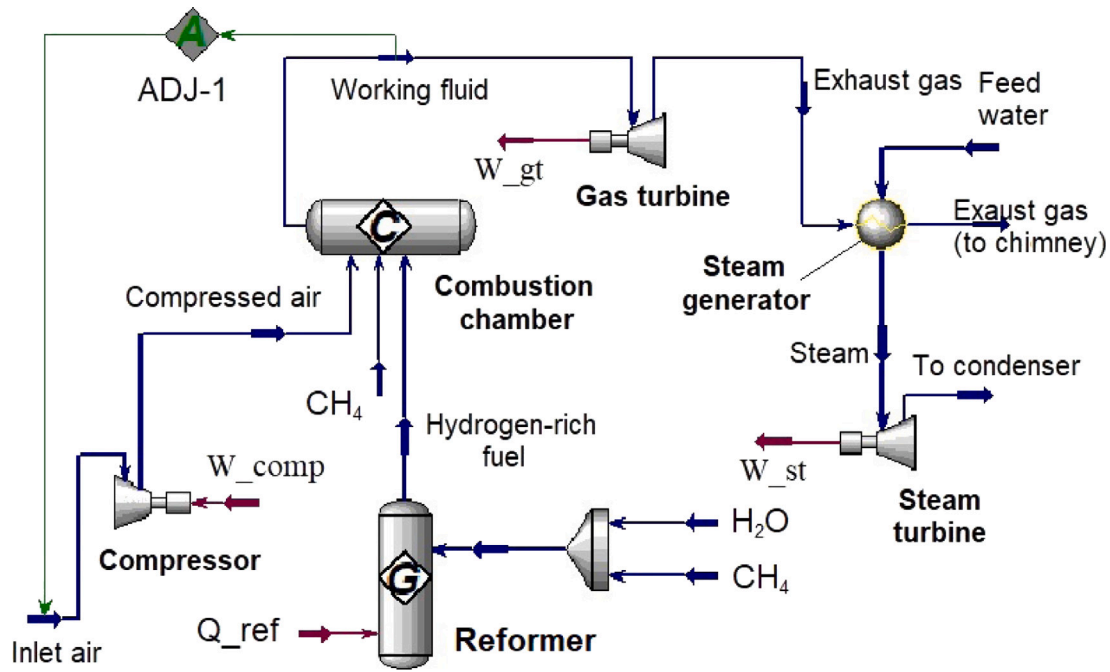
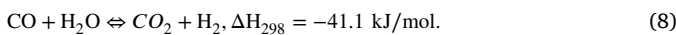
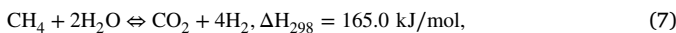
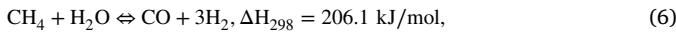


Fig. 5. A computational scheme of a combined cycle power plant with steam methane reforming and fired for hydrogen-rich fuel.

The above-discussed examples show that hydrocarbon fuel reforming before combustion can be considered a potential transition technology on the way to carbon-zero energy. In this technology, hydrogen-rich fuel is produced directly at the plant where it will be consumed. To understand the perspectives of such technology for a pathway to lower CO₂ emission in gas turbines, a carbon-zero energy source (e.g. solar energy) is used to compensate for the endothermic effect of the steam methane reforming process.

3.1. Methodology

The main concept of on-board hydrogen production technology is based on hydrocarbon fuel reforming before combustion in a gas turbine. In this study, the steam methane reforming process is considered. This process in the temperature range can be expressed by the following equations:



As can be seen, the reforming products contain a notable amount of hydrogen; therefore, the reforming products can be considered as hydrogen-rich fuel. The steam methane reforming process is endothermic and external heat should be supplied to accomplish this process.

In this paper, a combined cycle power plant with steam methane reforming before combustion in a gas turbine was analyzed. An analysis was performed for various cases determined by the different shares of initial fuel that reformed before combustion: from 0 to 100% of initial methane. The parameters of the gas turbine and steam turbine cycles are the same as for a gas turbine fueled by methane-hydrogen blends (Table 2).

A computational scheme of a combined cycle power plant with steam methane reforming and fired for hydrogen-rich fuel is presented in Fig. 5. In this scheme, the methane is partially reformed before combustion. The minimum share of the reformed CH₄ is 0% and the maximum share of the reformed CH₄ is 100%. The temperature of all inlet streams in the scheme is 20 °C.

The LHV and HHV efficiency of the combined cycle power plant with steam methane reforming is determined based on these expressions [103]:

$$\eta_{LHV} = \frac{N_{CCPP}}{\dot{m}_{\text{CH}_4} \cdot LHV_{\text{CH}_4} + Q_{ref}}; \quad (9)$$

$$\eta_{HHV} = \frac{N_{CCPP}}{\dot{m}_{\text{CH}_4} \cdot HHV_{\text{CH}_4} + D_{steam} \cdot 2264 + Q_{ref}};$$

where N_{CCPP} is power output of combined cycle power plant; $\dot{m}_{\text{CH}_4}$ is a mass of methane; LHV_{CH_4} , HHV_{CH_4} is lower and higher heating value of CH₄; D_{steam} is mass flow rate of steam for reforming; Q_{ref} is heat flow for the steam methane reforming process; 2264 is latent heat of H₂O at p=1 bar.

The composition of the reforming products and heat flow for reforming is depending on operational parameters: temperature, pressure, and inlet steam-to-methane ratio. Moreover, the catalyst type, residence time, reformer design, etc. have a significant effect on the reforming and flow characteristics. However, the thermodynamic analysis is considering only effect of temperature, pressure, and inlet steam-to-methane ratio. For more detailed analysis, an experimental investigation or numerical model using chemical kinetic models can be applied [128]. In this study, the thermodynamic analysis of the steam methane reforming process is performed via Gibbs free energy minimization method. This method is analyzed ideal conditions to describe the equilibrium state of the chemical system. The chemical system is in thermodynamically favorable condition when its total Gibbs free energy is at a minimum and its differential is essentially zero for the given operating conditions:

$$dG_{(t,p)} = 0 \quad (10)$$

where $G^{t,p}$ is total Gibbs free energy under specified temperature and pressure.

The overall principle of the Gibbs free energy minimization method has been described in our recent papers [129,130] and the papers of other authors [131].

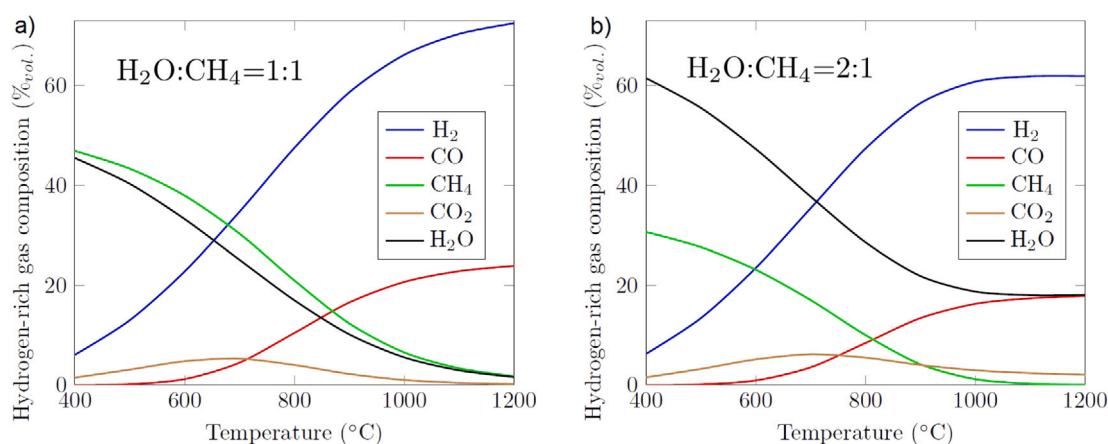
The final form of the equation for Gibbs free energy has the following view:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (11)$$

Table 3

The comparison of the synthesis gas composition at a steam-to-methane ratio of 2, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000 K	1 bar	HYSYS	63.55	14.22	16.11	5.22	0.9
		Verkhivker [85]	63.42	14.43	16.11	5.12	0.92
		Classical [132]	63.01	14.32	16.32	5.23	1.12
1000 K	5 bar	HYSYS	61.08	14.85	17.57	5.13	1.55
		Verkhivker [85]	61.44	14.93	17.97	4.18	1.48
		Classical [132]	60.66	14.82	17.32	5.55	1.65
1100 K	1 bar	HYSYS	63.78	15.95	16.2	3.98	0.09
		Verkhivker [85]	63.76	16.11	16.11	3.94	0.08
		Classical [132]	63.88	16.08	16.02	3.93	0.09
1100 K	5 bar	HYSYS	61.24	14.71	18.11	4.28	1.66
		Verkhivker [85]	61.45	14.93	17.96	4.18	1.48
		Classical [132]	61.28	15.02	17.87	4.24	1.59

**Fig. 6.** The composition of reforming products as a function of temperature for the different inlet steam-to-methane ratios of 1:1 and 2:1 at working pressure in a gas turbine of 21.5 bar.

where G_{fi}^0 is standard Gibbs free energy of i th component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure i th component; f_i^0 is fugacity of a pure i th component at the standard state. Peng–Robinson state equation is used in the thermodynamic analysis because the applicability range of this equation in terms of temperature and pressure is large. In addition, Peng–Robinson state equation is suitable for the systems containing hydrocarbons, steam, and other gases.

4. Results and discussion

The thermodynamic analysis of SMR was performed for a wide range of operating parameters. A comparison of the results of hydrogen-rich gas composition calculated in the present paper with the results obtained by other authors is presented in Table 3. The upper values are calculated via Aspen HYSYS in the present work, the middle rows are the mole fraction from an article published by Verkhivker and Kravchenko [85], the bottom results are the mole fraction of each component that was calculated based on the classical thermodynamic analysis [132]. A good correlation is observed for all results.

The steam methane reforming process is endothermic and this process is proceeding with an increase in the total mole number of the reforming products. Therefore, according to Le-Chatelier's principle, the reforming products is depending on temperature, pressure, and inlet steam-to-methane ratio. Fig. 6 shows the composition of reforming products as a function of temperature for the different inlet steam-to-methane ratios of 1:1 and 2:1 at working pressure in a gas turbine of 21.5 bar.

Fig. 6 shows that an increase in temperature leads to an increase in hydrogen and carbon monoxide mole fraction while methane and

steam mole fractions is decreasing. At a steam-to-methane ratio of 1:1, the products of reforming is mainly consisting of H₂ and CO in the temperature range above 1000 °C. The mole fraction of hydrogen in the products of reforming is more than 70% for H₂O:CH₄=1:1. Hydrogen becomes a dominant gas in the reformate in the temperature range above 680 °C for H₂O:CH₄=1:1. An increase in a steam-to-methane ratio leads to an increase in the mole fraction of H₂O in the products of reforming and a decrease in H₂ mole fraction. Fig. 6b shows that the maximum hydrogen mole fraction is about 60%, while H₂O mole fraction is about 20% for H₂O:CH₄=2:1. Hydrogen becomes a dominant gas in the reformate in the temperature range above 680 °C for H₂O:CH₄=2:1. Fig. 6 makes it clear that the products of steam methane reforming can be considered as hydrogen-rich fuel. One of the main advantages of this fuel is the possibility to use renewable heat sources for the methane reforming reaction.

The pressure is also affecting the hydrogen fraction in the hydrogen-rich fuel from the reforming process. Fig. 7 depicts the effect of pressure on H₂ volume and mass fraction for the different temperature levels at a steam-to-methane ratio of 1:1.

Fig. 7a shows that pressure has a notable effect on H₂ volume fraction, especially in the temperature range below 900 °C. H₂ volume fraction at 700 °C is 50.55% for 5 bar and 33.19% for 25 bar. Fig. 7b shows that the maximum H₂ mass fraction is about 17% in the temperature range above 1000 °C. Based on the results from Fig. 7 it can be concluded that a hydrogen-rich fuel obtained through steam methane reforming is similar (from point of view of carbon content) to a hydrogen-rich fuel obtained through methane dilution with hydrogen. In a hydrogen-rich fuel obtained through reforming, approx. 75% (vol.) of hydrogen is merged with approx. 23% of carbon monoxide (with 1 atom of carbon). In a hydrogen-rich fuel obtained through methane

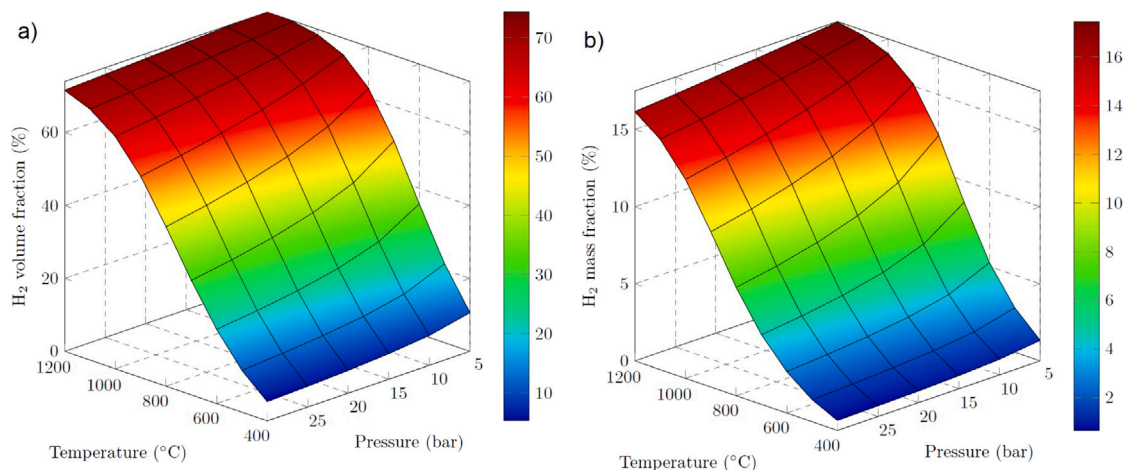


Fig. 7. Hydrogen volume and mass fraction vs. pressure and temperature for a steam-to-methane ratio of 1:1.

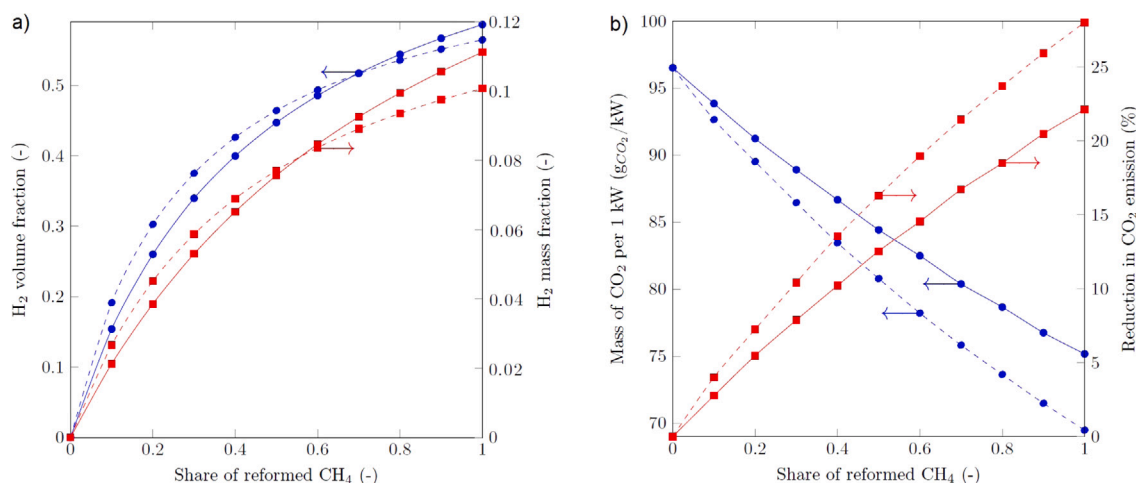


Fig. 8. Hydrogen volume and mass fraction in a hydrogen-rich fuel, and mass of CO_2 per 1 kW of a power output of a combined cycle power plant and a reduction in CO_2 emission as a function of a share of reformed methane for a steam-to-methane ratio of 1:1 (solid lines) and 2:1 (dashed lines) at $T=900^\circ\text{C}$ and $p=21.5$ bar.

dilution, 75% of hydrogen is merged with 25% of methane (with 1 atom of carbon). However, the combustion heat of carbon monoxide is about 3 times less than it of methane. This reduction in the combustion heat is compensated by heating a hydrogen-rich fuel in the reformer (according to a scheme from Fig. 5). To better understand the dependence of carbon dioxide emission on the different compositions of a hydrogen-rich fuel obtained via steam methane reforming, CO_2 emission, and mass flow rates are analyzed.

According to the schematic diagram from Fig. 5, hydrogen-rich fuel is obtained through reforming the partial or full amount of methane. The reforming temperature is 900°C and the pressure is 21.5 bar (pressure of a gas turbine). The temperature level of 900°C is achievable for the solar steam methane reforming process [133–135]. Fig. 8a shows a hydrogen volume and mass fraction in a hydrogen-rich fuel as a function of a share of reformed methane for a steam-to-methane ratio of 1:1 (solid lines) and 2:1 (dashed lines). Fig. 8b shows the mass of CO_2 per 1 kW of a power output of a combined cycle power plant and a reduction in CO_2 emission as a function of a share of reformed methane for a steam-to-methane ratio of 1:1 (solid lines) and 2:1 (dashed lines).

Fig. 8a illustrates that reforming of 20% of initial methane gives a hydrogen-rich fuel with about 30% of hydrogen volume fraction. When 100% of methane is reformed, the hydrogen-rich fuel contains more than 60% (vol.) and 11% (mass) of hydrogen. Fig. 8a shows the effect of a steam-to-methane ratio on hydrogen yield. An increase in a steam-to-methane ratio leads to an increase in hydrogen yield per 1 mol of CH_4 .

However, when 100% of methane is reformed, the share of hydrogen is decreasing due to a high residual H_2O in the products of reforming.

Fig. 8b shows the effect of methane reforming on CO_2 emission. When 100% of methane is reformed, $69.47 \text{ g}_{\text{CO}_2}/\text{kW}$ and $75.17 \text{ g}_{\text{CO}_2}/\text{kW}$ for a steam-to-methane ratio of 1:1 and 2:1, respectively, is observed. This level of CO_2 per 1 kW of power output is equivalent to a hydrogen-rich fuel obtained through methane dilution with hydrogen up to 50% (vol). An increase in a steam-to-methane ratio leads to a decrease in CO_2 emission because in this case, the enthalpy of the reforming process is higher, and more renewable heat can be converted into electricity without CO_2 emission.

An increase in the share of H_2O in a steam–methane mixture leads not only to an increase in H_2 yield but also to an increase in heat for steam generation and heating of reacting steam–methane mixture. This fact is a reason for efficiency decreasing. Fig. 9a shows the effect of LHV and HHV efficiency as a function of the share of reformed CH_4 . Fig. 4a illustrates that an increase in the share of reformed CH_4 is a reason for decreasing in LHV and HHV efficiency. This decrease can be explained by the fact that with an increase in the share of reformed CH_4 the heat for steam generation is increasing. Moreover, Fig. 9a shows that HHV efficiency is decreasing faster than LHV efficiency. This can be explained by the fact the combustion products contain additional H_2O from the steam methane reforming process. H_2O volume fraction is presented on the right side of Fig. 9. Therefore, the same issue on the addition of the condensing heat exchangers after a steam generator

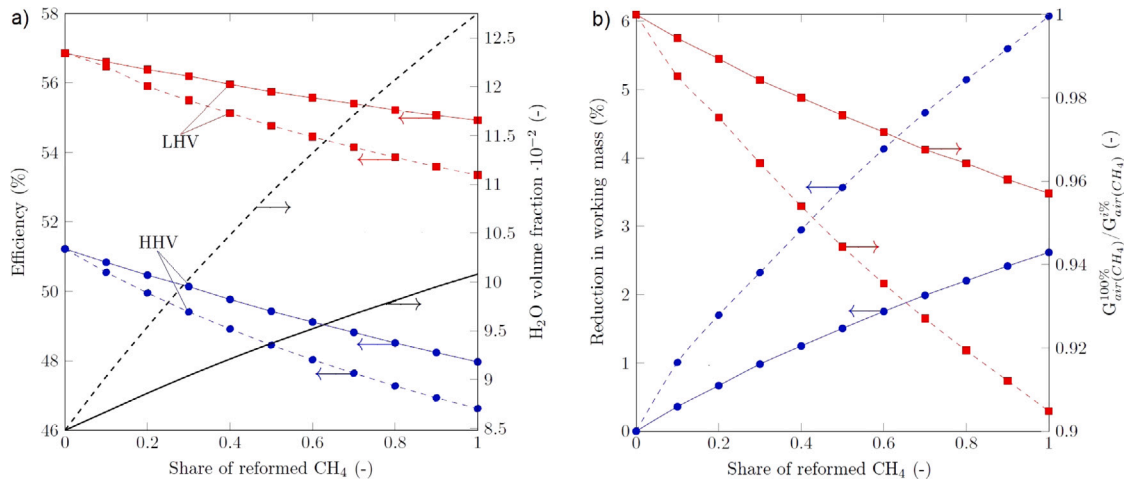


Fig. 9. LHV and HHV efficiency of a combined cycle power plant fired with a hydrogen-rich fuel and H₂O volume fraction in a working fluid (a) and reduction in mass flow rate of working fluid and air (b) as a function of the share of reformed CH₄.

as for hydrogen-rich fuel obtained through methane dilution with H₂ is observed.

To understand the effect of utilization of a hydrogen-rich fuel obtained via steam methane reforming, the mass flows through a gas turbine are analyzed. Fig. 9b that a reduction in working gas mass flow rate is about 2% and 6% for a steam-to-methane ratio of 1:1 and 2:1, respectively. In this case, the decrease in an air mass flow rate is compensated by an increased fuel mass flow rate due to the addition of H₂O in fuel. The reduction in working gas mass flow rate up to 2% for H₂O:CH₄=1:1 is almost the same as in a hydrogen-rich fuel obtained via methane dilution with hydrogen up to 60% (vol.). Therefore, steam methane reforming will have no notable effect on flow parameters in a turbine part. However, steam methane reforming before combustion leads to a notable effect on an air mass flow rate through a compressor. The reforming of 100% of methane leads to a decrease in an air mass flow rate of 4.2% and 9.8% for H₂O:CH₄=1:1 and H₂O:CH₄=2:1, respectively. A decrease in an air mass flow rate is a reason of decreasing in compressor power, and as a result, is a reason of an increase in the overall gas turbine output power.

5. Conclusion

The crucial conclusions of this study are summarized below:

- Using hydrogen-rich gas as a fuel is one of the potential ways to reduce CO₂ emission in the gas turbines without a significant modification of the existing power generation plant. Two main ways of hydrogen-rich fuel generation can be highlighted: natural gas dilution with hydrogen; on-board hydrogen-rich gas production via hydrocarbon fuel reforming;
- Different physical and calorimetric properties of hydrogen and methane lead to a non-linear effect of hydrogen volume fraction on the reduction of CO₂ emission for hydrogen-rich fuel obtained via H₂ addition: 20% (vol) of H₂ gives a reduction in CO₂ emission of 7.2%; 50% (vol) of H₂ gives 23.5% reduction; 75% of H₂ gives 51.1% reduction;
- Hydrogen addition to methane leads to an insignificant effect on LHV efficiency; however, an increase in H₂ volume fraction leads to a decrease in HHV efficiency due to an increase in a volume fraction of H₂O in a working fluid. In comparison with 100% CH₄, HHV efficiency reduction is 0.7% for H₂ 50%; 1.3% for H₂ 75%; and 1.8% for H₂ 100%;
- Increase in H₂ volume fraction in hydrogen-rich fuel leads to a decrease in a fuel mass flow rate: 56400 kg/h for H₂=0%; 48400 kg/h for H₂=50% (16.8% less than for H₂=0%); 22920 kg/h for H₂=100% (60.4% less than for H₂=0%). However, the effect on a mass flow rate of a working fluid is insignificant. An increase in H₂ volume fraction from 0 to 50% leads to a decrease in a mass flow rate of the working fluid of 1.7%; from 0 to 100% of 6.4%;
- Hydrogen-rich fuel for utilization in the gas turbines can be produced via steam methane reforming with a renewable heat source. The effect of temperature, pressure, and inlet steam-to-methane ratio is determined and the maximum hydrogen volume fraction is about 70%. When 20% of initial methane is reformed a hydrogen-rich fuel with about 30% of hydrogen volume fraction is produced. When 100% of methane is reformed, the hydrogen-rich fuel contains more than 60% (vol.) and 11% (mass) of hydrogen;
- When 100% of methane is reformed, 69.47 gCO₂/kW and 75.17 gCO₂/kW for a steam-to-methane ratio of 1:1 and 2:1, respectively, is observed. This level of CO₂ per 1 kW of power output is equivalent to a hydrogen-rich fuel obtained through methane dilution with hydrogen up to 50% (vol); the maximum reduction in CO₂ emission is about 25% for hydrogen-rich fuel produced via steam methane reforming;
- An increase in the share of reformed CH₄ is a reason for decreasing LHV and HHV efficiency due to a higher volume fraction of H₂O in a working fluid up to 2.5% and 4.8%, respectively, for H₂O:CH₄=2:1; 1.7% and 2.4%, respectively, for H₂O:CH₄=1:1; a reduction in working gas mass flow rate is about 2% and 6% for a steam-to-methane ratio of 1:1 and 2:1, respectively;
- hydrogen-rich fuel produced via H₂ dilution and via steam methane reforming can be considered as a pathway to a significant reduction in CO₂ emission on the existing power plants based on the gas turbines. In these cases, the mass flows through a gas turbine will be slightly changed with a low decrease in LHV and HHV efficiency. Therefore, the schematic diagrams of such power plants will not need a notable modernization, only fuel supply infrastructure will be changed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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